Assignment # 4

STAT 679: Statistical Methods for Spatial Lattice Data–Spring 2026, UW-Madison

In this homework assignment, we will be working with the Environmental Quality Index (EQI) data set from the Environmental Protection Agency (EPA). The environmental quality index provides a snapshot of environmental quality across the United States at the county level with data collected from 2006–2010. For this assignment, we will focus on Illinois and Wisconsin. The variables in this dataset, `enviro_ilwi.shp`, are:

- `State`: The state, "WI" or "IL".
- `Cnty_Nm`: County name.
- `GEOID`: FIPS code, where the first two numbers indicate the state and last three numbers indicate the county.
- `socio`: An index of socio-demographic factors. This includes factors such as percent unemployment, median household income, percent of families living at less than the poverty level, amongst others.
- `built`: An index of built environment factors. This includes factors such as education and healthcare-related environment, construction, housing, and many other built environment factors.
- `EQI`: An index of environmental quality factors.

For index variables `socio`, `built`, `EQI`, lower (more negative) values represent "better" quality and higher (more positive) values represent "worse" quality. For example, as `EQI` gets more negative, the better the environmental quality. As they get more positive, the worse the environmental quality is.

For more detailed information, please reference the Environmental Quality Index Technical Report (2006–2010).

For this homework assignment, we will also introduce the `sf` package. There are two ways to work with spatial data in R. So far, we have shown you how to work with `SpatialPolygonsDataFrames`. Another approach is to use the `sf` package and to work with a spatial dataframe of class `sf`.

The dataset `enviro_ilwi.shp` can be loaded into R in the following way:

```
library(sf)         # for loading in the shapefile for this new sf approach
library(spatialreg) # for the SAR/CAR spatial regression functions
library(spdep)      # for spatial nbd & list of spatial weights
library(ggplot2)    # for visualization
library(tidyr)      # for data manipulation

# load in the shape file using sf approach
enviro <- st_read("enviro_ilwi.shp")
class(enviro)  # this will say "sf" and "data.frame"
```

1. **10 pts** Perform an exploratory data analysis and exploratory spatial data analysis. Be sure to include a table of summary statistics, at least 1 visualization, and at least 1 map. The response variable (y) is `EQI` and predictor variables (x's) are `socio` and `built`.

   Hint: You may find the code below useful:

   ```
   # making a scatter plot
   p1 <- ggplot(data=dat, aes(x=var, y=EQI)) +
         geom_point(alpha=0.5) + theme_bw() + geom_smooth()

   # making a map
   ggplot() +
       geom_sf(data=dat, aes(fill=var)) +
       coord_sf() +
       scale_fill_viridis_c() +
       theme_bw() +
       xlab("Longitude") +
       ylab("Latitude")
   ```

2. **10 pts** Perform Moran's I test (under randomization) for `socio` and `built` using binary spatial weights in the list of spatial weights. Interpret your findings.

   Hint: You may find the code below useful:

   ```
   nb <- poly2nb(dat, row.names=dat$GEOID)
   listB <- nb2listw(nb, style="B")
   set.seed(1)
   moran.test(dat$x, listB, alternative="two.sided")
   ```

3. **10 pts** (refer to *Poverty in Wisconsin and Illinois Data Example – Simple Linear Regression*). Build a standard linear regression model for `EQI` as the response variable and `socio`, `built` as the predictor variables. Be sure to perform diagnostics of the model by residual plots.

4. **10 pts** (refer to *Poverty in Wisconsin and Illinois Data Example – Spatial Linear Models*) Build a SAR and CAR model using `EQI` as the response variable and both `socio`, `built` as the predictor variables, again using binary spatial weights. Complete the interpretation for all regression coefficients in both the SAR and CAR model. Test both the SAR and CAR model residuals for spatial correlation. Which model do you prefer?